Detailed Step-by-Step Working Plan

Proactive State-Aware and Thermal-Resilient Inference Scheduling for Sustained Edge Vision Systems

April 14, 2026

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Roles and Responsibility Split

Student Responsibilities: Implementation, coding, telemetry pipeline, scheduler development, experiments, data collection, plotting, and draft writing.

PI Responsibilities: Research direction, paper framing, contribution clarity, review of methodology, validation of claims, final writing refinement, and submission strategy.

Joint Responsibilities: Experiment design decisions, interpretation of results, and final paper positioning.

Purpose of This Plan

This document gives the graduate student a detailed, paper-ordered working plan for turning the current project into a complete journal-style systems paper. The plan follows the structure of the manuscript section by section, so that technical implementation, experiments, analysis, writing, and artifact preparation all progress in the same order that the final paper will be written.

This plan also marks what has already been completed. Those items should not be redone unless quality control shows a problem. Instead, the student should verify them, preserve reproducibility, and build the remaining work on top of them.

Project Status Summary

At the time of this plan, the following major items have **already been completed**:

- The problem motivation has been established: sustained YOLOv8n inference on Raspberry Pi 5 under passive cooling reaches thermal throttling.
- The dataset has already been prepared and partitioned reproducibly.
- Annotation conversion and visual verification have already been completed.
- The YOLOv8n baseline model has already been trained.
- Validation metrics, confusion matrix, and training curves have already been generated.
- A worst-case thermal benchmark has already been designed and run.
- A progress report and reviewer feedback have already been produced.

The **major missing elements** are:

- Full implementation of the proactive scheduler.
- Implementation and evaluation of the High-Confidence Confirmation (HCC) mechanism.
- Full comparative evaluation against baselines.
- Ablation studies.
- Final paper rewriting from progress-report form into full journal-paper form.

1 Overall Working Rules for the Student

Before describing the section-by-section plan, the student should follow these rules throughout the project:

1.1 Reproducibility Rule

Every experiment must be reproducible. Every run should save:

- date and time,
- git commit hash or version label,
- hardware setup,
- software environment,
- configuration parameters,
- random seeds,
- raw telemetry logs,
- inference outputs,
- summary metrics.

1.2 No Silent Changes Rule

Do not silently change model weights, data partitions, metric definitions, or scheduler logic. Every important change must be logged in a short change log.

1.3 Paper-First Rule

All code, experiments, and plots should be created with the final paper in mind. If a result will not help a figure, table, claim, or appendix, do not spend too much time on it.

1.4 Baseline Freeze Rule

The already trained YOLOv8n model should remain the default frozen baseline unless the PI explicitly approves retraining. The purpose of the next phase is scheduler evaluation, not model redesign.

Section-by-Section Working Plan Following the Paper Structure

2 Title, Abstract, and Contribution Framing

This section comes first in the paper, but should be finalized last. However, the student should begin maintaining the material needed for it immediately.

2.1 Task 1: Maintain a Live Contribution List

Create a small text file called `contributions_live.md`. Update it every week with three categories:

1. What is already proven.
2. What is partially implemented.
3. What is not yet validated.

Already done:

- Baseline reproducible training pipeline.
- Thermal throttling problem demonstrated empirically.

Still needed:

- Final validated contribution statement after scheduler experiments.

2.2 Task 2: Maintain a Claim-vs-Evidence Table

Create a two-column internal table:

- Column 1: paper claim,
- Column 2: exact figure/table/experiment that supports it.

This will prevent overclaiming.

Already done:

- Some claims about dataset preparation, training reproducibility, and thermal vulnerability are already supported.

Still needed:

- Support for all scheduler and HCC claims.

3 Introduction

The introduction in the final paper must present the problem, explain why it matters, define the gap, and state the paper's contributions clearly.

3.1 Task 3: Rewrite the Introduction as a Full Systems-Paper Introduction

The student should draft a new introduction with this order:

1. Importance of sustained edge inference.
2. Why thermal throttling and instability matter in long-duration deployments.
3. Why static optimization is insufficient.
4. Why reactive thresholding is insufficient.
5. What this paper contributes.

Already done:

- Core motivation has already been written in the progress report.
- The thermal problem has already been demonstrated experimentally.

Still needed:

- Rewrite the section so it no longer reads like a progress report.
- Remove speculative language about scheduler effectiveness until results exist.
- Add a concise contribution paragraph at the end of the introduction.

3.2 Task 4: Draft the Final Contribution Paragraph

At the end of the introduction, the student should prepare a 3-item contribution list in prose form. It should eventually look like:

1. a proactive state-aware scheduler using derivative-based telemetry,
2. a bounded-cost HCC mechanism,
3. a long-horizon experimental evaluation on Raspberry Pi 5.

Already done:

- The conceptual contribution is already known.

Still needed:

- Do not finalize the wording until experiments are complete.

4 Related Work

The review specifically asked for broader engagement beyond Raspberry Pi-specific studies. This section must therefore be strengthened.

4.1 Task 5: Expand the Literature Matrix

Create a spreadsheet or table with the following columns:

- citation,
- problem domain,
- platform,
- proactive or reactive,
- thermal-aware or not,
- online adaptation or static,
- what limitation remains.

Already done:

- A useful initial literature review already exists in the progress report.

Still needed:

- Add broader works on dynamic inference scaling.
- Add adaptive model selection literature.
- Add edge systems/control literature relevant to scheduling.
- Add recent thermal-aware runtime management papers beyond Raspberry Pi case studies.

4.2 Task 6: Reorganize Related Work into Subthemes

The student should rewrite related work under four subsections:

1. thermal limits in edge inference,
2. static model optimization for edge deployment,
3. dynamic and runtime-adaptive inference,
4. thermal-aware scheduling and system control.

Already done:

- Many relevant citations have already been identified.

Still needed:

- Reframe the literature around the exact gap the paper addresses.
- End the section with a gap paragraph that clearly states what is missing in prior work.

5 Problem Definition and Research Questions

The final paper needs a clearer formal problem statement than the progress report currently has.

5.1 Task 7: Write the Formal Problem Definition

The student should define:

- what the system observes,
- what it controls,
- what failure or stress events are,
- what objective is optimized.

The final text should define stress events such as:

- thermal throttling,
- voltage sag or instability,
- frame-rate collapse,
- unacceptable accuracy degradation.

Already done:

- The broad problem is already described in proposal material and revision drafts.

Still needed:

- Convert the intuition into a formal section in the manuscript.

5.2 Task 8: State the Research Questions Explicitly

Write 3–4 research questions, for example:

1. Can derivative-based telemetry predict impending thermal stress early enough to prevent throttle events?

2. Can proactive scheduling improve long-horizon FPS stability relative to reactive baselines?
3. What accuracy-energy-stability tradeoff is achieved by the scheduler?
4. Does HCC recover critical detections with limited overhead?

Already done:

- These questions are already implicit in the proposal and review analysis.

Still needed:

- Write them explicitly in the paper and align experiments to them.

6 System Design

This is the most important unfinished section. It must explain the scheduler precisely and in enough detail that another researcher could reproduce it.

6.1 Task 9 (Student-led, PI-reviewed): Define the Configuration Space

The student should document all allowed inference configurations:

$$c(t) = \{r, p, f\}$$

where:

- r is resolution,
- p is precision,
- f is frame-rate cap.

For each option, the student should write down:

- exact values tested,
- whether it is already implemented,
- any unsupported combinations.

Already done:

- The idea of the configuration space is already defined conceptually.

Still needed:

- Implement all candidate runtime switching options in code.
- Verify all configurations run stably on the Pi.

6.2 Task 10 (Student-led): Implement the Telemetry Pipeline

The student must build a synchronized telemetry logger collecting:

- temperature $T(t)$,
- temperature derivative $\dot{T}(t)$,
- utilization $U(t)$,

- utilization derivative $\dot{U}(t)$,
- voltage $V(t)$,
- voltage derivative $\dot{V}(t)$,
- optionally current and power.

For each signal, document:

- data source,
- sampling rate,
- smoothing method,
- synchronization method with inference events.

Already done:

- Thermal benchmarking has already shown that the platform can be profiled meaningfully.

Still needed:

- Build a unified logging pipeline for all signals.
- Derive and validate slope estimates.
- Ensure timestamp alignment between telemetry and inference outputs.

6.3 Task 11 (Student-led): Implement the Scheduler State Representation

Write and code the system state:

$$s(t) = [T(t), \dot{T}(t), U(t), \dot{U}(t), V(t), \dot{V}(t)].$$

The student should:

1. define exact units,
2. choose smoothing windows,
3. choose derivative calculation method,
4. test signal stability under idle and stress conditions.

Already done:

- The state vector has already been specified conceptually in the proposal/revision material.

Still needed:

- Convert it into a fully implemented software component.

6.4 Task 12 (Student-led, PI-reviewed): Implement the Decision Policy

The student should implement the scheduler objective, such as

$$c^*(t) = \arg\min_{c \in \mathcal{C}} \left(\alpha(t)E(c) + \beta(t)(1 - A(c)) + \Phi(c, t) \right).$$

This requires:

- a lookup or estimate for energy cost $E(c)$,
- a lookup or estimate for expected accuracy $A(c)$,
- a risk penalty $\Phi(c, t)$,
- adaptation rules for $\alpha(t)$ and $\beta(t)$.

The student should start with a simple heuristic version and then refine it if needed.

Already done:

- The mathematical idea has already been drafted in proposal materials.

Still needed:

- Implement the policy in code.
- Add logging that records why each action was chosen.
- Add safety constraints and hysteresis so switching does not oscillate.

6.5 Task 13 (Student-led): Implement High-Confidence Confirmation (HCC)

The student should define:

- what triggers HCC,
- what higher-fidelity configuration is used during HCC,
- how many frames are used,
- how the event is logged,
- how the additional energy cost is measured.

Already done:

- HCC has already been proposed conceptually.

Still needed:

- Implement HCC in code.
- Define trigger logic precisely.
- Evaluate whether HCC improves relative detection quality under constrained execution.

6.6 Task 14: Produce a System Overview Figure

Create a clean system diagram showing:

- input stream,
- telemetry logging,
- state construction,
- scheduler decision,
- inference configuration,
- HCC path,

- outputs and logs.

Already done:

- No finalized system figure yet.

Still needed:

- Make this figure early, because it will help both writing and implementation consistency.

7 Experimental Setup

Some of this section is already complete and should be stabilized rather than reinvented.

7.1 Task 15: Freeze and Verify Dataset Preparation

The student should verify:

- the exact dataset subset used,
- the split counts,
- the partition script,
- the seed values,
- the annotation conversion script.

Already done:

- Dataset partitioning has already been completed reproducibly.
- Annotation conversion and visual verification have already been completed.

Still needed:

- Package all scripts and counts neatly for the appendix or artifact bundle.

7.2 Task 16: Freeze and Verify Model Training Details

The student should create a single source of truth for:

- architecture,
- epoch counts,
- best epoch,
- optimizer,
- augmentation settings,
- final weights file,
- validation metrics.

Already done:

- YOLOv8n training has already been completed.
- Validation metrics, confusion matrix, and training curves already exist.

Still needed:

- Ensure the trained weights used in scheduler experiments are fixed and archived.

7.3 Task 17: Define the Input Workloads for Scheduler Evaluation

The student should define three workload modes:

1. worst-case stress workload,
2. moderate workload,
3. realistic deployment workload.

For each workload, specify:

- source (stitched test video, live stream, curated clip),
- duration,
- repeatability,
- why it is included.

Already done:

- A worst-case benchmark stream has already been created and justified.

Still needed:

- Add at least one or two additional scenarios so the scheduler is not evaluated only on an extreme stress case.

7.4 Task 18: Define All Baselines Before Running Main Experiments

The student must implement and document at least:

- Static-Max,
- Static-Min,
- reactive thermal-threshold scheduler,
- simple utilization-based reactive scheduler,
- proposed scheduler,
- proposed scheduler + HCC.

Already done:

- The need for these baselines is already established from the work plan and review.

Still needed:

- Implement all baselines in the same software framework.
- Ensure they all log outputs in the same format.

7.5 Task 19: Lock the Metric Definitions

The student must define exact computation for:

- mean FPS,
- FPS variance and percentiles,
- time-to-throttle,

- time above thermal thresholds,
- energy per inference,
- energy per unit time,
- accuracy proxy or mAP,
- stress-event count,
- HCC trigger count and cost.

Already done:

- Some baseline metrics have already been reported for model quality and thermal behavior.

Still needed:

- Finalize one metrics document and never change it without PI approval.

8 Results

This section will become the core of the final paper. The student should build it in a staged order.

8.1 Task 20 (Student-led): Produce Configuration Profiling Results

Before main scheduler evaluation, profile all runtime configurations. For each configuration, measure:

- throughput,
- average temperature rise rate,
- average power,
- energy per inference,
- relative detection quality.

Already done:

- Not yet done completely.

Still needed:

- Generate the lookup tables needed by the scheduler and the paper.

8.2 Task 21 (Student-led): Run Short Pilot Scheduler Tests

Before long experiments, run short tests of 5–15 minutes to verify:

- scheduler decisions are sensible,
- telemetry reacts as expected,
- switching does not oscillate,
- logs are complete,
- HCC triggers correctly.

Already done:

- Not yet done.

Still needed:

- Complete this before any paper-quality long-run experiments.

8.3 Task 22 (Student-led): Run the Main Long-Horizon Experiments

The student should run repeated trials for all methods across all workload scenarios. For each condition:

- use a fixed duration such as 60–120 minutes,
- run at least 5 repetitions if feasible,
- keep ambient conditions documented,
- save all raw logs.

Already done:

- Passive-cooling thermal vulnerability has already been demonstrated.

Still needed:

- Run the full comparison matrix across baselines and proposed methods.

8.4 Task 23 (Student-led, PI-reviewed): Build the Main Results Tables and Figures

The student should prepare:

- a table of average metrics across methods,
- a time-series plot showing temperature, FPS, and configuration over time,
- a plot of time-to-throttle across methods,
- an energy-vs-accuracy or stability tradeoff figure.

Already done:

- Training curves and confusion matrix already exist for the model section.

Still needed:

- All scheduler-focused results figures still need to be generated.

8.5 Task 24 (Student-led, PI-reviewed): Add Statistical Analysis

For each major result:

- compute mean and standard deviation,
- compute confidence intervals where appropriate,
- report percentage improvement over baselines,
- note any unstable or inconsistent conditions.

Already done:

- Not yet done for scheduler results.

Still needed:

- Add formal statistical reporting so the paper looks complete and rigorous.

9 Ablation Studies

The ablations are not optional.

9.1 Task 25: Telemetry Ablation

Run versions of the scheduler using:

- temperature only,
- temperature + utilization,
- temperature + voltage,
- full telemetry set.

Already done:

- Not yet done.

Still needed:

- Use identical workloads and durations to ensure fair comparison.

9.2 Task 26: Derivative Ablation

Compare:

- absolute telemetry only,
- absolute + derivative telemetry.

Already done:

- Not yet done.

Still needed:

- This is important because derivative-based anticipation is central to novelty.

9.3 Task 27: HCC Ablation

Compare:

- scheduler without HCC,
- scheduler with HCC.

Measure:

- extra energy cost,
- effect on accuracy proxy,
- effect on stability.

Already done:

- Not yet done.

Still needed:

- Produce one table and one figure showing whether HCC is worthwhile.

9.4 Task 28: Policy Stability Ablation

Test what happens if hysteresis or dwell-time safeguards are removed.

Already done:

- Not yet done.

Still needed:

- This helps show why the final scheduler design choices are necessary.

10 Discussion

The discussion must interpret results honestly and connect them back to the problem and literature.

10.1 Task 29 (Joint, PI-led interpretation): Write the Practical Interpretation of Results

Once experiments are complete, the student should explain:

- when the scheduler helps most,
- when it helps less,
- what tradeoffs are being made,
- whether HCC meaningfully protects accuracy,
- what deployment environments benefit most.

Already done:

- Some practical motivation is already present in the progress report.

Still needed:

- Replace speculative discussion with evidence-based interpretation.

10.2 Task 30 (PI-led, Student drafts): Write a Dedicated Limitations Subsection

The paper should openly discuss:

- Raspberry Pi 5 specificity,
- dependence on available telemetry,
- limitations of the chosen dataset,
- class imbalance, especially D40,
- possible mismatch between stitched-image benchmark and real video deployments,
- heuristic rather than learned policy design, if that remains true.

Already done:

- The D40 issue has already been identified in both the report and the review.

Still needed:

- Expand limitations into a mature discussion section rather than a defensive note.

11 Conclusion

The conclusion should become short, evidence-based, and non-speculative.

11.1 Task 31 (PI-led, Student drafts): Rewrite the Conclusion Only After Final Results

The student should write the conclusion in this order:

1. restate the problem,
2. summarize what was built,
3. summarize the strongest evidence,
4. state the main takeaway,
5. briefly mention future work.

Already done:

- A progress-report conclusion already exists.

Still needed:

- Rewrite it completely once the scheduler results are available.

12 Artifacts, Appendix, and Internal Review Preparation

12.1 Task 32: Build the Artifact Package

Prepare:

- all scripts,
- environment file,
- run instructions,
- raw and processed results organization,
- plotting scripts,
- README.

Already done:

- Reproducibility practices are already strong in dataset and training setup.

Still needed:

- Extend this rigor to scheduler code and long-run experiments.

12.2 Task 33: Prepare an Internal PI Review Package

Before submission, the student should prepare a package containing:

- the near-final manuscript,
- all figures and tables,
- a one-page summary of claims,
- a list of unresolved weaknesses,
- a run log index,
- a mapping from reviewer concerns to paper changes.

Already done:

- Prior review comments already exist and can be used as a checklist.

Still needed:

- Build the final pre-submission review packet.

13 Suggested Week-by-Week Execution Order

The student should follow the order below. This order mirrors the paper structure but is also practical for implementation.

Phase 1: Stabilize Existing Work

1. Verify dataset scripts, partition files, and annotation conversion.
2. Verify final model weights and archive them.
3. Collect all existing plots and metrics in one organized folder.

Phase 2: Build the System Design Components

1. Implement telemetry collection.
2. Implement state estimation and derivatives.
3. Implement configuration switching.
4. Implement scheduler logic.
5. Implement HCC.
6. Make the system overview figure.

Phase 3: Prepare the Evaluation Framework

1. Define workloads.
2. define baselines,
3. lock metric definitions,

4. build analysis scripts,
5. run short pilot tests.

Phase 4: Produce Main Results

1. profile all configurations,
2. run long-horizon experiments,
3. generate tables and figures,
4. compute statistics.

Phase 5: Produce Ablations

1. telemetry ablation,
2. derivative ablation,
3. HCC ablation,
4. policy stability ablation.

Phase 6: Finish Writing

1. rewrite introduction,
2. strengthen related work,
3. write system design,
4. write experimental setup using already completed materials,
5. add results and ablations,
6. write discussion and limitations,
7. rewrite conclusion.

Phase 7: Final Packaging

1. build artifact package,
2. run internal review,
3. revise paper,
4. prepare submission files.

14 Definition of Completion

The student's work is complete when all of the following are true:

- The scheduler and HCC are implemented and reproducible.

- Baselines are implemented and compared fairly.
- Main long-horizon results are complete.
- Ablation studies are complete.
- All figures and tables are finalized.
- The manuscript reads as a complete journal paper rather than a progress report.
- Every major claim in the paper is linked to a result.
- The PI can review one organized artifact folder and reproduce the main findings.

Final Note to the Student

Do not think of this as "just implementing a scheduler." The real task is to build a complete and defensible systems paper. Every technical decision should therefore be made with two questions in mind:

1. Can this be justified scientifically?
2. Can this be explained clearly in the paper?

If the answer to either question is no, stop and revise the approach before proceeding further.